

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Winter 2023

Course: B. Tech

Branch: Electrical & Allied Engineering

Semester: IV

Subject Code & Name: Analog and Digital Electronics (BTBS404)

Max Marks: 60

Date: 23-01-24

Duration: 3 Hrs.

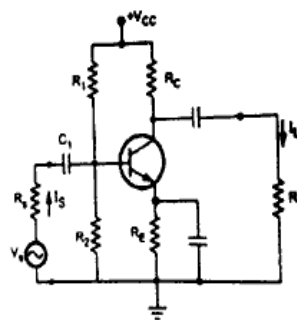
Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

Marks

Q.1 Solve Any Two of the following.

- A) Explain the principle of operation of transistor as an amplifier. **06**
- B) Compare CB, CC and CE amplifier. **06**
- C) Consider a single stage CE amplifier with $R_s = 1 \text{ k}\Omega$, $R_1 = 50 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$, $R_c = 1 \text{ k}\Omega$, $R_L = 1.2 \text{ k}\Omega$, $h_{fe} = 50$, $h_{ie} = 1.1 \text{ k}\Omega$, $h_{oe} = 25 \text{ }\mu\text{A/V}$ and $h_{re} = 2.5 \times 10^{-4}$, as shown in fig. Find A_i , R_i , A_v , A_{is} , A_{vs} and A_p . **06**



Q.2 Solve Any Two of the following.

- A) What are various characteristics of an ideal Op-Amp? Discuss in details. **06**
- B) Draw the block diagram of an Op-amp and explain the purpose of using each block. **06**
- C) Explain Op-Amp as a differentiator with neat circuit diagram and necessary equations. **06**

Q.3 Solve Any Two of the following.

- A) Convert the following numbers: **06**
- a) $(111.101)_2 = (\quad ? \quad)_{10}$
 - b) $(378.93)_{10} = (\quad ? \quad)_8$
 - c) $(A0F9.0EB)_{16} = (\quad ? \quad)_{10}$
- B) Solve the following arithmetic operations **06**
- a) Subtract $(5)_{10}$ from $(7)_{10}$ using 2's complement method of binary subtraction.
 - b) Multiply 1001 by 1101.
 - c) Divide 1110101 by 1001.
- C) Which gates are called Universal gates? Derive NOT, AND and OR gates from NAND gate. **06**

Q.4 Solve Any Two of the following.

- A) What are different operational properties of logic family? Explain any 3 properties in detail. 06
- B) Explain 2-input RTL NOR gate. 06
- C) Explain J-K flip-flop with circuit diagram and truth table. 06

Q. 5 Solve Any Two of the following.

- A) Reduce the expression for $f(A, B, C, D) = \Sigma_m(1, 5, 6, 12, 13, 14) + d(2, 4)$ using K-map. 06
- B) Explain half adder circuit with the truth table. 06
- C) Reduce the expression for $f(A, B, C, D) = \pi M(2, 3, 4, 5, 6, 7, 8, 11, 12)$ 06

*** End ***